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17 September 1982
SUPERSEDING
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USED IN LIEU OF
MIL-D-16680B
19 February 1954
(See 6.4)

MILITARY SPECIFICATION

DECK COVERING MAGNESIA AGGREGATE MIXTURE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an asbestos-free deck covering used in galleys, pantries, and butcher shops aboard ship. It should not be used for areas or locations where it is subjected to continual wetting, such as washrooms, water closet spaces, and under equipment where deck is generally wet. It is for single-coat application.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standard (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-P-143 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.
UU-S-48 - Sacks, Shipping Paper.
PPP-B-35 - Bags, Textile, Shipping, Burlap, Cotton and Waterproof Laminated.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-D-723 - Drums, Fiber.
PPP-D-760 - Drums and Pails, Metal (5 and 16.64 Gallon).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5610

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MILITARY

MIL-S-901 - Shock Tests, H. I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-D-3135 - Deck Covering Underlay Materials.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-147 - Palletized Unit Loads 40" x 48" Pallets.
MIL-STD-1623 - Fire Performance Requirements and Approved Specification for Interior Finish Materials and Furnishings (Naval Shipboard Use).

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. Deck-covering furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein and has been listed on or approved for listing on the applicable qualified products list.

3.2 Material. The deck-covering shall consist essentially of an intimate mixture of asbestos-free dry ingredients and a gaging solution of magnesium chloride proportioned to form a covering which will be suitable for application with a trowel. A certificate of compliance shall be required that the mixture is asbestos-free (see 6.1.1). The ingredients shall be capable of being mixed and laid to the specified thickness in accordance with good commercial practice and shall be capable of producing wearing surfaces which are smooth, dense, and relatively free from waves, trowel marks, depressions, and other surface defects.

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3.2.1 The dry ingredients proportioned on a weight basis, shall be intimately premixed, and shall consist of finely ground caustic calcined magnesia; asbestos-free inert aggregates such as silica, talc, expanded vermiculite, marble chips, granulated cork, and inorganic pigment.

3.2.2 Gaging solution. The gaging solution, of 21.5° ± 0.5° Baume, shall be made from magnesium chloride flakes dissolved in tap water, to which may be added a stabilizer, if necessary.

3.2.3 Color. Color shall be terra cotta, unless otherwise specified (see 6.1).

3.3 The covering shall be capable of adhering to the deck or structure on which applied without the use of clips or other devices welded to the deck, or other reinforcement not a part of the compound as mixed for application.

3.4 Resistance to staining. The covering shall be capable of being readily cleaned at frequent intervals without appreciable loss of surface or color, and shall not be adversely affected by detergents when tested as specified in 4.5.2.

3.5 Density. When tested as specified in 4.5.3, the covering, after drying, shall not exceed 3.00 pounds per square foot for a thickness of 1/4 inch.

3.6 Resistance to impact. When tested as specified in 4.5.4, the covering shall show no visible signs of chipping, cracking, or detachment from the steel plate. There shall be not more than 1/16 inch of permanent indentation.

3.7 Indentation. When tested as specified in 4.5.5, the initial indentation of the deck-covering shall be not more than 0.003 inch. The covering shall show no signs of cracking or becoming detached from the steel plate as a result of the indentation.

3.8 Resistance to elevated temperature. The covering shall not flow or slip in any part more than 1/16 inch, nor soften, when tested as specified in 4.5.6.

3.9 Resistance to moisture and temperature changes. The covering shall show no signs of cracking, separation from the steel plate, or other failure when tested as specified in 4.5.7.

3.10 Moisture absorption. The covering shall not absorb more than 7.0 percent of water, based on its weight at normal atmospheric conditions, when tested as specified in 4.5.8.

3.11 Nonslip properties. The covering shall show factors of friction not less than that shown in table I when tested as specified in 4.5.9.

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TABLE I. Nonslip properties.

	Contacting surface condition		
	Dry	Wet	Oily
Factor of static friction:			
Leather	0.30	0.50	----
Rubber	.60	.70	0.50
Factor of sliding friction:			
Leather	.30	.40	----
Rubber	.50	.80	.30

3.12 Resistance to wear. The covering shall show wear not to exceed 0.150 inch, when tested as specified in 4.5.10.

3.13 Fire resistance. The deck system shall conform to the fire test requirement set forth in MIL-STD-1623 (see 4.5.11).

3.14 Shear strength.

3.14.1 The initial shear strength of the covering shall be not less than 50 pounds per square inch when tested as specified in 4.5.12.

3.14.2 The shear strength of the covering after oxygen bomb aging shall be not less than 85 percent of the initial shear strength when tested as specified in 4.5.12.

3.15 Resistance to corrosion. The covering shall not soften or become detached and the surface of the steel beneath the covering shall show no signs of corrosion when tested as specified in 4.5.13.

3.16 Shock resistance. The covering shall show no signs of chipping, cracking, or detachment from the steel plate when tested as specified in 4.5.14.

3.17 Directions for application. Each container shall be clearly labeled with instructions for application of the covering.

3.18 Serviceability. The deck-covering shall show no breaks, loss of adhesion, or other deficiency when examined during and after the minimum service test specified in 4.5.15.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the

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inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to the Naval Sea Systems Command. These tests shall consist of the tests specified in 4.5.

4.2.1 A sample consisting of sufficient premixed dry ingredients (finish coat) and magnesium chloride flakes for the application of 5 square feet approximately 1/4 inch thick shall be submitted for testing.

4.3 Sampling.

4.3.1 Lot. For the purposes of sampling, a lot shall consist of one production batch or run.

4.3.2 Sampling for examination of filled containers. A random sample of filled containers shall be selected in accordance with MIL-STD-105 at inspection level I and acceptable quality level = 2.5 percent defective to verify compliance with this specification in regard to fill, closure, and marking. Each type of container shall be sampled separately.

4.3.3 Sampling for lot acceptance tests. From each lot offered for delivery, samples shall be selected in accordance with table II for the tests specified in 4.4.3. The material for each sample shall be selected from a different container. Each sample shall consist of premixed dry ingredients (finish coat) and magnesium chloride flakes.

TABLE II. Sampling for tests.

Lot size: Pounds of the dry mixed ingredients	Number of samples to be selected for the tests of 4.5.3, 4.5.7, and 4.5.12	Number of samples to be selected for the tests of 4.5.4 and 4.5.10
Up to 2000	1	1
2001 to 8000	2	2
8001 to 22,000	3	2
22,001 and over	5	2

4.4 Examination and tests.

4.4.1 Examination of filled containers. Each sample filled container selected as specified in 4.3.2 shall be examined for defects of construction of the containers and the closure, for evidence of leakage, and for unsatisfactory markings; each filled container shall also be weighed to determine the amount of contents. Any filled container in the sample having one or more defects, or under required fill, shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected.

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4.4.2 The samples selected in accordance with table II shall be mixed in accordance with the manufacturer's instructions and subjected to the appropriate tests specified in 4.4.3. If a sample is found to be not in conformance with this specification, the entire lot shall be rejected.

4.4.3 Lot acceptance tests. Lot acceptance tests shall consist of the tests specified in 4.5.3, 4.5.4, 4.5.5, 4.5.7 and 4.5.12 (initial shear strength only).

4.4.3.1 Small lots. Tests shall be omitted when the lot consists of 2,000 pounds or less.

4.5 Test procedures.

4.5.1 Preparation of specimens. Unless otherwise specified, specimens of the sizes as specified in the following tests shall be made by preparing the covering in accordance with the manufacturer's instructions and troweling it onto 1/8-inch thick clean mild steel plates to a thickness of approximately 1/4-inch by means of wooden templates. The specimens shall then be cured for 96 hours at room temperature before conducting tests. Except as specified hereinafter all tests shall be conducted at a temperature of 70° to 75° Fahrenheit (°F) and a relative humidity of 65 plus or minus 2 percent.

4.5.2 Resistance to staining. A specimen 6 by 2 inches shall be immersed for 1 hour in the solution specified herein so that a 3-inch section of the specimen is immersed in the solution and the upper half remains unexposed for purposes of comparison. The liquid solution shall be made by dissolving soap powder in tepid water to form a 1/2 of 1 percent solution, then adding lamp black until the solution is dark gray. After immersion, the specimen shall be rinsed for 1 minute with tap water at 140°F and dried with an air jet. The covering shall then be examined for softening, stains, streaking, or loss of color.

4.5.3 Density. The deck-covering shall be applied to three 6-inch-square 1/8-inch-thick mild steel plates which have been previously measured and weighed. When the material has dried for 168 hours, the three test specimens, including the steel plates, shall each be weighed to the nearest 0.1 gram (gm). The length and width shall be measured to the nearest 0.1 inch, and the thickness to the nearest 0.001 inch. From the difference between the weight of the covered steel plate and the uncovered steel plate, the density of the material shall be computed in ounces per square foot per the thickness as applied. The density shall be the average of the three specimens. The thickness shall be determined by measuring the steel plates, with and without the covering, at 16 equally distributed points on the specimen, by means of a dial thickness gage and a template. The difference in thickness of the steel plate and the covered steel plate shall be averaged to determine the thickness of the material.

4.5.4 Resistance to impact. Two specimens 6 inches square shall be used. Each specimen shall be tested separately after being firmly held on a solid-horizontal base. A 2-pound steel ball shall be dropped vertically from a height of 8 feet on the deck covering so that the impact will be at the center of the specimen. Each specimen shall be subjected to two impacts of the ball.

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4.5.5 Indentation. One specimen 6 inches square shall be used. Three indentations shall be made on the covering and the results averaged. The center of each indentation shall be not less than 1-1/2 inches from the edge of the specimen, and not less than 2-1/2 inches from the center of the adjoining indentation.

4.5.5.1 A load of 2,000 pounds shall be applied on the deck-covering for 30 minutes by means of a circular indenter with a cross-sectional area of 1 square inch and a radius of 1/64 inch on the perimeter of the indenting flat face. The thickness of the deck covering shall be measured with a micrometer dial gage, using a 4-ounce weight and 1/4-inch diameter flat foot on both sides of the specimen before and after the 2,000-pound load has been applied.

4.5.5.2 The initial indentation shall be taken as the difference between the thickness of the material before indentation and immediately after the load has been removed.

4.5.6 Resistance to elevated temperature. The resistance of the material to elevated temperature shall be determined as follows:

4.5.6.1 Flow or slip. A specimen 6 by 2 inches shall be scribed with a line parallel to and approximately 1 inch from a 2-inch edge used as a reference. The distance between this line and the edge of the steel plate shall be measured to the nearest 0.01 inch. The specimen shall then be suspended vertically from the end opposite the reference and in an oven maintained at a constant temperature of 158°F for 5 hours. When the specimen has cooled to room temperature, the distance between the reference edge and the line shall be measured again. The difference between the two measurements is called the flow or slip.

4.5.6.2 Softening. The deck-covering shall be examined by touch, immediately after the specimen has been removed from the oven, to determine whether the material has softened under the action of heat.

4.5.7 Resistance to moisture and temperature changes. Two specimens 6 by 2 inches shall be immersed in a solution of 4 percent sodium chloride in water, under a pressure of 8 pounds per square inch, for 48 hours. Immediately after immersion the specimens shall be examined to determine whether the covering has softened or become detached and then quickly subjected to two complete cycles of alternate exposure to a temperature of 0° ± 5°F, for 24 hours, followed by a temperature of 120° ± 5°F for 24 hours. The specimens shall then be examined for evidence of cracking or other failure.

4.5.8 Moisture absorption. Three specimens 2 inches square by 1/4-inch thick shall be prepared by applying the deck-covering to oiled surfaces of steel plate, in such a manner that upon drying the specimens will not adhere to the plates. Each specimen without the steel backing plate shall be weighed dry, dipped into tap water at room temperature, lightly wiped on all surfaces with a paper towel and again weighed to the nearest 0.1 gm. Immediately after weighing, the specimen shall again be immersed in the tap water for 24 hours, lightly wiped and again weighed. The percent gain in moisture shall be based on the weight of the dry specimen and the difference between the weight after 24-hour immersion and the weight after dipping and wiping.

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4.5.9 Nonslip properties. The factors of friction of the deck-covering against leather and rubber respectively, shall be determined. The leather shall be oak-tanned sole leather which has been sanded smooth with grade 0 garnet paper. The rubber shall be a vulcanized compound having a shore A durometer hardness of 60 to 80. Tests shall be made with the contact surface dry, wet with a solution of 4 percent sodium chloride in water and oiled, using SAE10W oil. A load of 22 pounds shall be applied uniformly over the 2- by 4-inch specimen. The determination of static and sliding friction may be made by the inclined plane method or by determining the pull on a spring balance required to start the test piece from rest and then to maintain a constant velocity. The factor of friction shall be the average of four readings.

4.5.10 Resistance to wear. Specimens 3 inches long by 2 inches wide shall be prepared so that an area of approximately 6 square inches is presented to the abrasive in the machine specified in 4.5.10.1. The abrasive grit shall be No. 80 aluminum oxide. Before wear-testing, the thickness shall be measured at 12 equally distributed points on the specimen by means of a dial thckness gage and a template. The specimen shall then be subjected to 1,500 revolutions of the wear test machine, and the thickness obtained at the same points previously measured. The difference in thickness shall be averaged to determie the thickness of material worn away by the abrasive.

4.5.10.1 Wear-test machine. The wear-test machine shown on figures 1 and 2 shall have the following essential mechanical characteristics:

(a) Mild steel, abrasion disk, 14-1/8 inches in diameter and 1/2 inch thick, revolved clockwise at a constant speed of 23.5 revolutions per minute (rpm).
(b) Specimen holder, 2 by 4 inches, attached to 5/8-inch diameter shaft by means of a universal coupling. The distance between centers of the disk and the specimen holder shaft is 5.1 inches. A specimen of decking material is secured in the specimen holder by means of clamps at each end of the holder. The holder revolves clockwise at a constant speed of 32.5 rpm.
(c) Hardened tool-steel cam attached to the specimen holder shaft by means of set screws. In each 1/2 revolution of the shaft, the specimen is lifted a distance of 1/16 inch and then dropped back on disk (a) under a 10-pound weight. Changes in the lift due to wear of the specimen is compensated by adjustment of the cam at regular intervals, so that the lift is maintained at a practically constant distance.
(d) Weight 10 pounds, the maximum found suitable for use with this test machine.
(e) Distributor for abrasive grits. The grits flow by gravity from the hopper (f) through a 13/64-inch inside diameter tube and into the center of the distribution chamber. An impeller in the chamber provides an evenly distributed feed of the grits through eight 3/64-inch diameter holes centered on a 1-5/8-inch diameter pitch circle. A uniform distribution of grits falls to disk (a) in the path of the test specimen. The distributor is revolved at a constant speed of 9 rpm.

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(f) Hopper for feeding abrasive grits.
(g) Bevel drive gear.
(h) Predetermining revolution counter and switch.
(i) One-third horsepower compound wound direct current motor.
(j) Hopper for collecting used abrasive grit.
(k) Steel frame.

Details of the distributor for the abrasive grit, the cam and the specimen holder are shown on figures 3 and 4, respectively.

4.5.10.2 Procedure. The steel disk 4.5.10.1(a) revolves while the weighed specimen of deck-covering in holder 4.5.10.1(b) is lifted and dropped by a cam while it is being revolved. When the specimen is dropped through motion of the cam, it falls a distance of 1/16 inch and meets disk 4.5.10.1(a) with impact simulating the fall of a foot on the deck surface. Foot action is further simulated in that the revolving specimen contacts the disk with a sliding and twisting motion until lifted clear by the cam. Continuous rotation of the specimen introduces the element of wear in all directions on the decking material. Free abrasive grits of No. 80 aluminum oxide are fed at a steady rate from the hopper 4.5.10.1(f) and distributed uniformly on the steel disk in the path of the specimen. The revolving motions of the steel disk and the test specimen keep the abrasive grits circulating to the outer side of the path of the specimen. The grits fall from the edge of the steel disk into hopper 4.5.10.1(j), from which they are collected and then screened to eliminate undesired particles before replacement in the upper feeding hopper 4.5.10.1(f).

4.5.11 Fire-resistance test. The completed deck covering system shall be tested in accordance with the test procedure set forth in MIL-STD-1623.

4.5.12 Shear strength. Twelve 2 by 6 inch specimens shall be prepared as specified in 4.5.1 except that the covering shall be applied only to a 2 inch square on the 2 by 6 inch steel. The 2 inch square shall be applied such that 1 inch of the steel plate is exposed at one end and 3 inches exposed at the opposite end. Specimens shall be tested by measuring the load required to shear the 2-inch square area of deck-covering from the steel plate by means of a shear test jig as shown on figure 5. Six specimens shall be tested to determine the initial shear strength. Six specimens shall be tested after aging in an oxygen bomb for 96 hours under a pressure of 300 pounds per square inch and temperature of 70° Centigrade (C)(158°F).

4.5.13 Resistance to corrosion. Two specimens 6 by 12 inches shall be prepared using a surface wetting coat of MIL-D-3135, type I (1 part latex paste, 2 parts underlay powder, by weight) to help adhere the covering to the steel plate. The specimens shall be immersed in a 10 percent salt (NaCl) solution for 15 days, during which tie a continuous stream of air shall be passed through the solution in order to promote corrosion. The specimens shall then be examined to determine whether the material has softened or become detached from the steel backing plates. A portion of the covering shall be removed carefully from the steel to observe any sign of rusting or corrosion of the steel plate beneath the bond coat.

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4.5.14 Shock resistance. Three 6 by 6 inch specimens shall be prepared as specified in 4.5.1 except that the covering shall be applied centrally to each of three 8 by 8-inch steel plates by means of a wooden template in such a manner that a 1-inch wide portion of the steel is exposed along the periphery of the specimens. Each specimen shall be subjected to HI(high impact) shock in a testing machine as specified in MIL-S-901. Each specimen shall be centrally secured to the test plate of the testing machine by eight 1/4-inch diameter machine screws, equally located along the periphery of the specimen plate 1/2-inch away from the edge of the specimen steel plate. Each specimen shall then be subjected to a series of shocks consisting of consecutive blows of 100, 200, 400, 700, 1,000, 1,400, and 2,000 foot-pounds (ft-lb) to provide approximately uniform increases of striking velocity. The 2,000-ft-lb blow shall then be immediately followed by a second 2,000-ft-lb blow. The specimens shall then be examined for chipping, cracking or detachment from the steel backing plate.

4.5.15 Serviceability. The deck-covering shall be applied aboard ship for a six-month service test. During this time period and at its completion, the deck covering shall be examined for breaks, loss of adhesion, or other deficiencies which would limit its serviceability.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Surface dressing. The surface dressing shall be furnished in 1-gallon cans or 5-gallon pails as specified (see 6.1).

5.1.1.1.1 Cans. The one-gallon cans shall conform to type V, class 2 of PPP-C-96. Exterior Plan B coating and side seam striping shall be required. Cans shall be provided with wire handles which shall be galvanized or protectively coated to resist corrosion.

5.1.1.1.2 Pails. The five-gallon pails shall conform to type II of PPP-D-760. Wire handles or bails shall be galvanized or protectively coated to resist corrosion.

5.1.1.2 Dry ingredients. No packaging will be required. Material shall be packed in containers as specified in 5.2.2.

5.1.2 Level C. The material shall be packaged in accordance with the suppliers commercial practice.

5.2 Packing. Packing shall be Level A, B, or C, as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Surface dressing. The cans and pails shall be packed in accordane with the Level A packing requirements of PPP-C-96.

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5.2.1.2 Dry ingredients. The dry ingredients shall be packed in any of the following containers. Selection of the container shall be at the option of the contractor. Gross weight of shipping container shall not exceed 100-pounds.

Specification	Container	Type or grade
UU-S-48 PPP-B-35 PPP-D-723	Sack, paper Bags Drums, fiber	Grade X Type III Type II grade A

5.2.1.2.1 Drums shall be provided with a barrier, lining or bag type liner to protect contents from moisture and sifting. Container closure shall be as specified in the applicable container specification or appendix thereto.

5.2.2 Level B.

5.2.2.1 Surface dressing. The cans and pails shall be packed in accordance with the Level B packing requirements of PPP-C-96.

5.2.2.2 Dry ingredients. The dry ingredients shall be packed in any of the following containers. Selection of the container shall be at the option of the contractor. Gross weight of shipping container shall not exceed 100-pounds.

Specification	Container	Type or grade
UU-S-48 PPP-B-35 PPP-D-723	Sack, paper Bags Drums, fiber	Grade L/W Type I or II Type I or II, grade A

5.2.2.2.1 Drums shall be provided with a barrier lining, or bag type liner to protect contents from moisture and sifting. Container closure shall be as specified in the applicable container specification or appendix thereto.

5.2.3 Level C. Material shall be packed in a manner which will insure acceptance by common carrier and safe delivery at destination. shipping containers or method of packing shall comply to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.3 Palletizations. When specified (see 6.1) pails, sacks, bags or drums, as applicable, shall be palletized in unit loads in accordance with MIL-STD-147.

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5.4 Marking. Shipment markings shall be in accordance with TT-P-143.

5.4.1 Special marking. Shipping containers shall be marked "ASBESTOS-FREE" (see 6.1).

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Color required (see 3.2.3).
(c) The deck-covering ingredients should be purchased by net weight.
(d) Level of packaging and packing required (see 5.1 and 5.2).
(e) Quantity required (see 5.1.1.1).
(f) Palletization, if required (see 5.3).
(g) Special markings required (see 5.4.1).

6.1.1 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.2	Certificate of compliance	DI-E-2121	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.1.1.1 The data requirements of 6.1.1 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.2 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List QPL-16680 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and

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manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.2.1).

6.2.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Navy - SH
Army - ME

Preparing activity:
Navy - SH
(Project 5610-0237)

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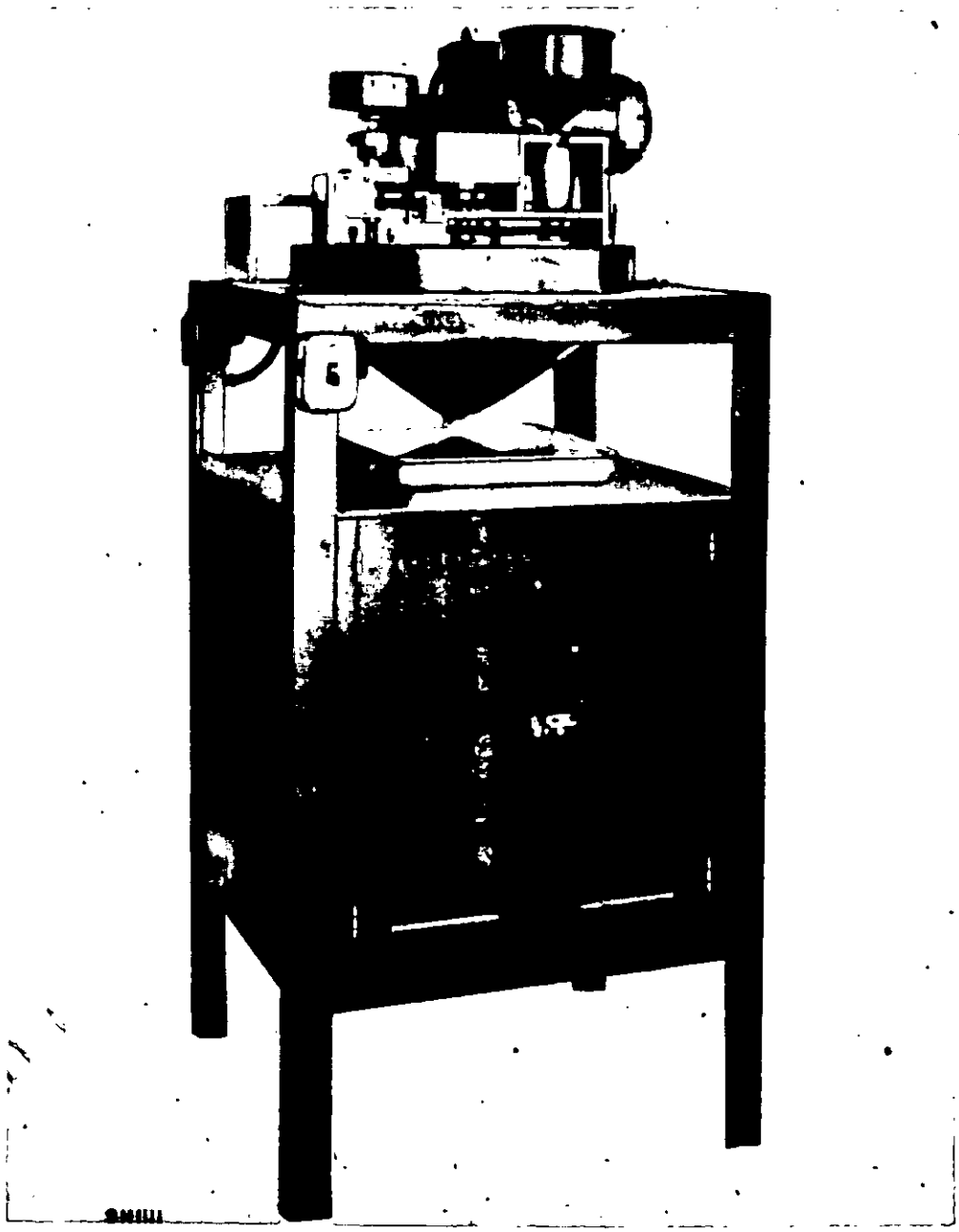


FIGURE 1. Wear test machine.

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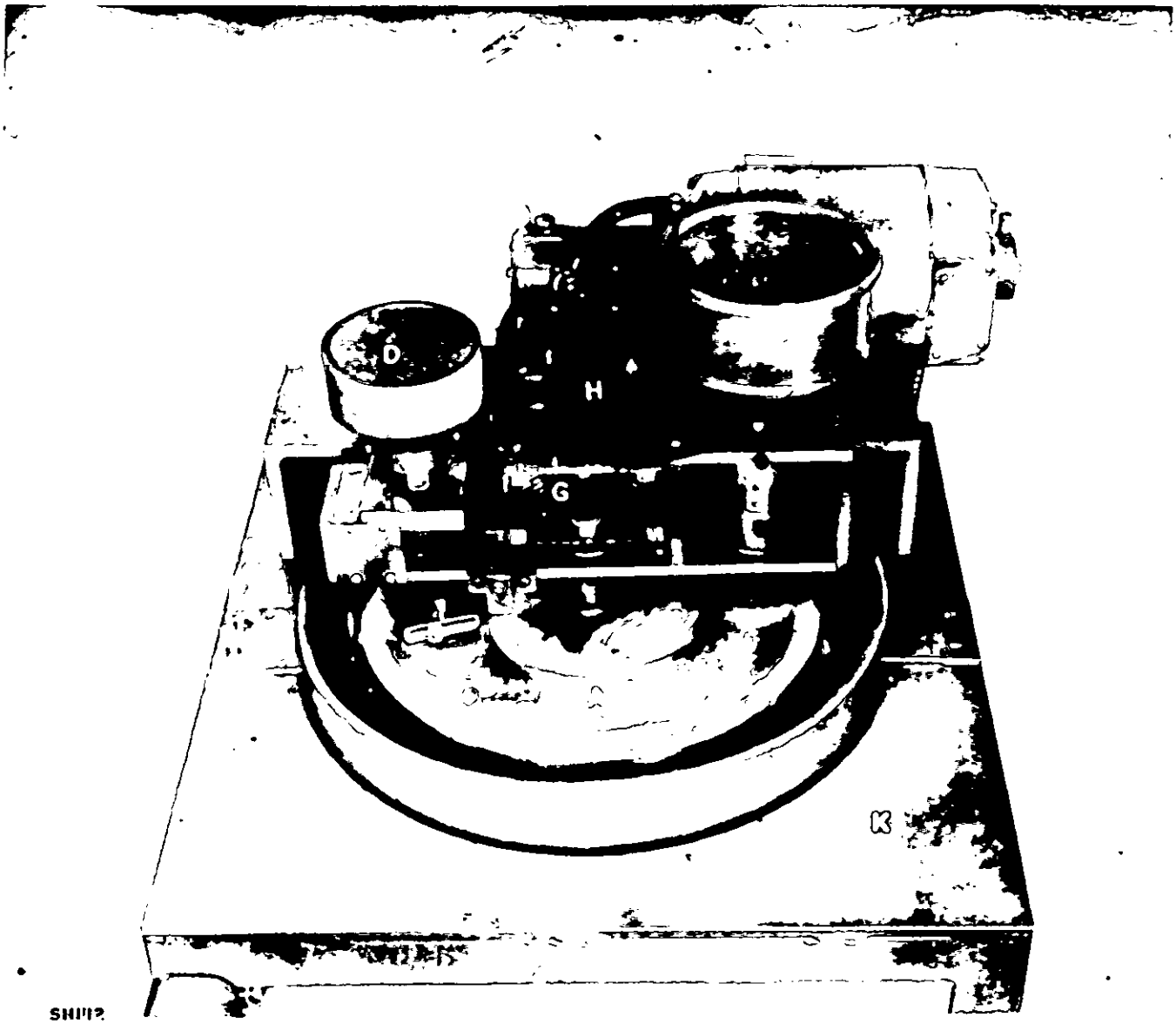


FIGURE 2. Detail view of wear test machine.

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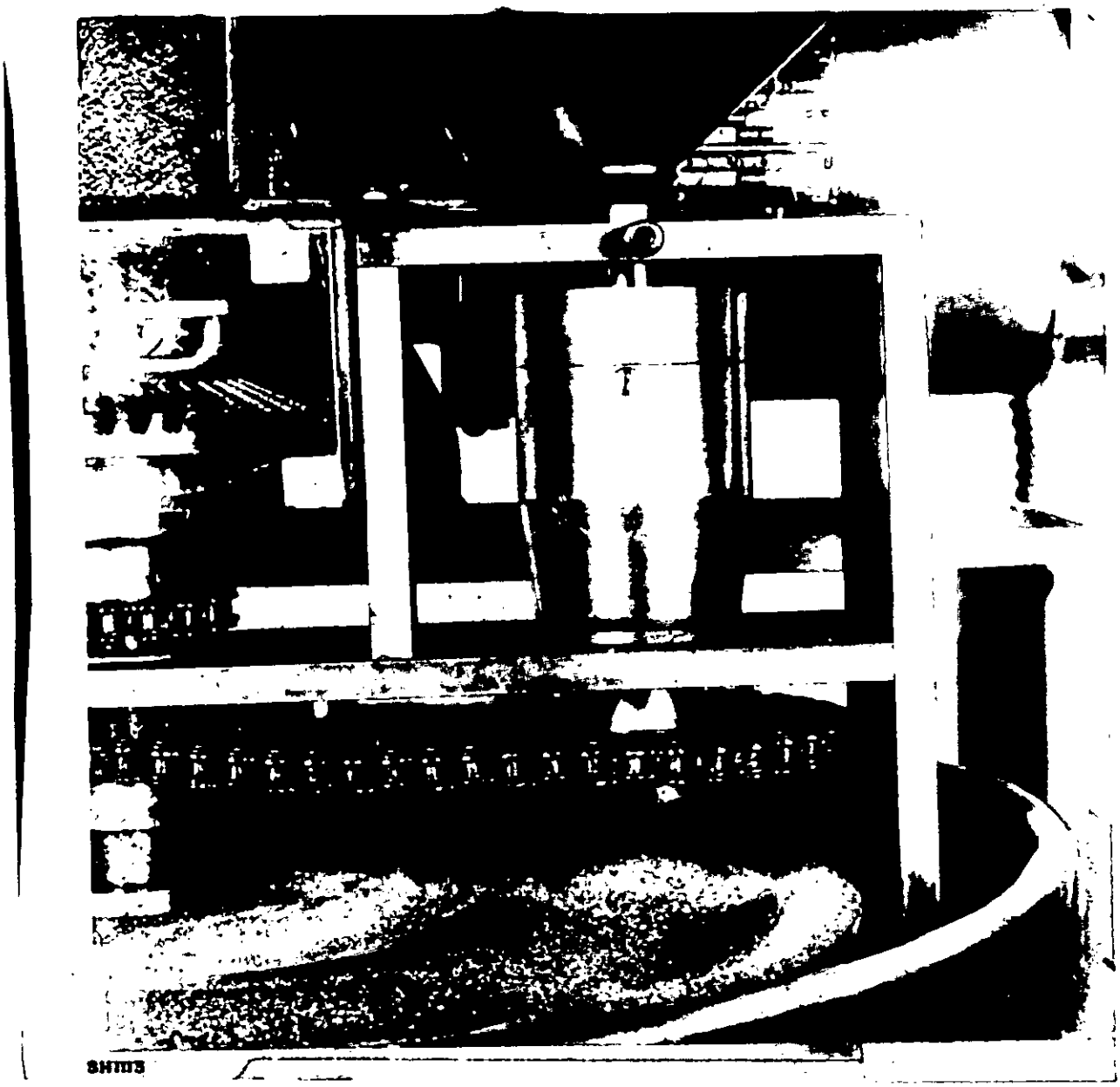
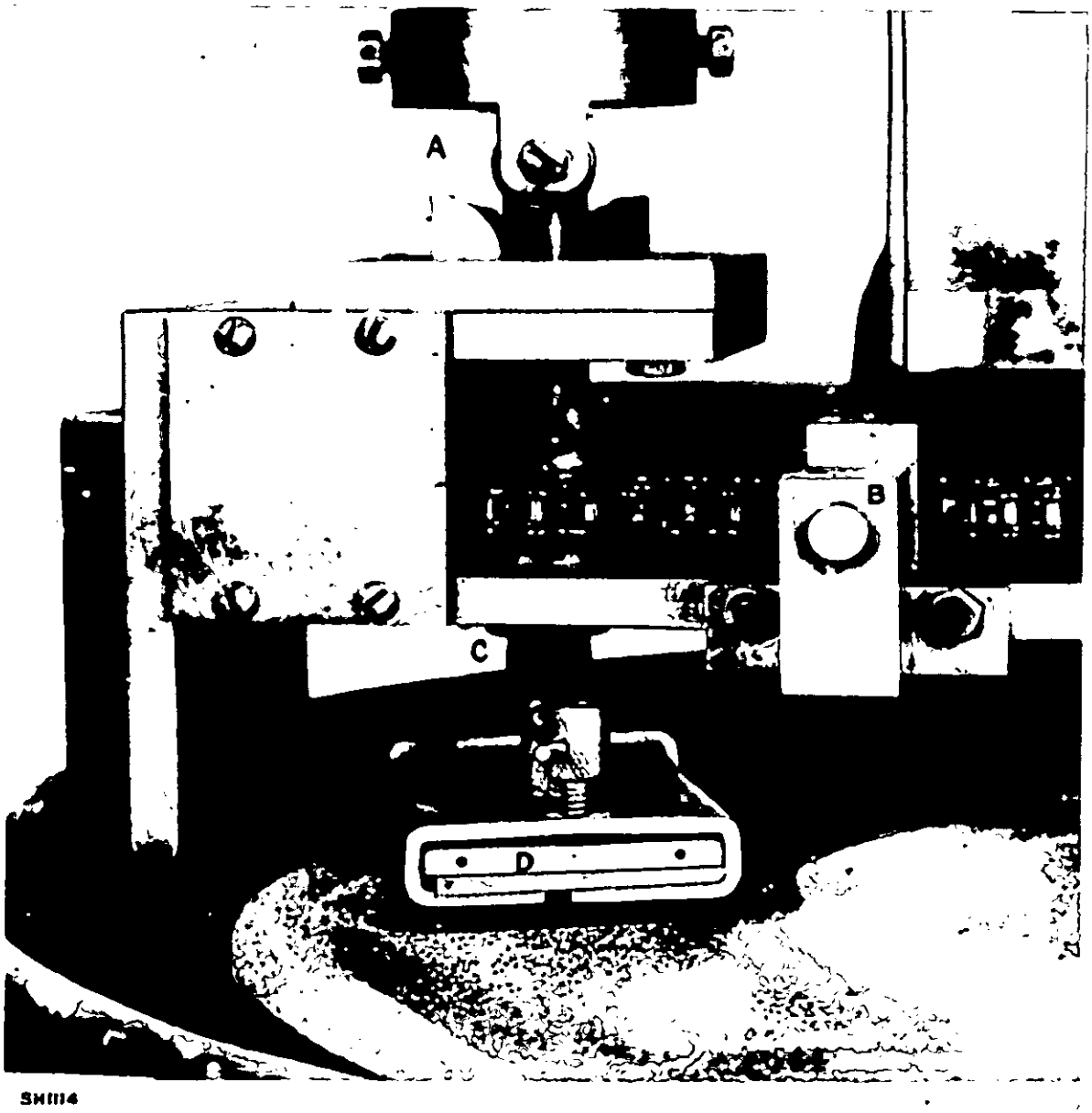


FIGURE 3. Detail of distributor of abrasive grits.

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A-Cam.
B-Chain idler.
C-Universal joint.
D-Specimen holder.

FIGURE 4. Detail of cam and specimen holder.

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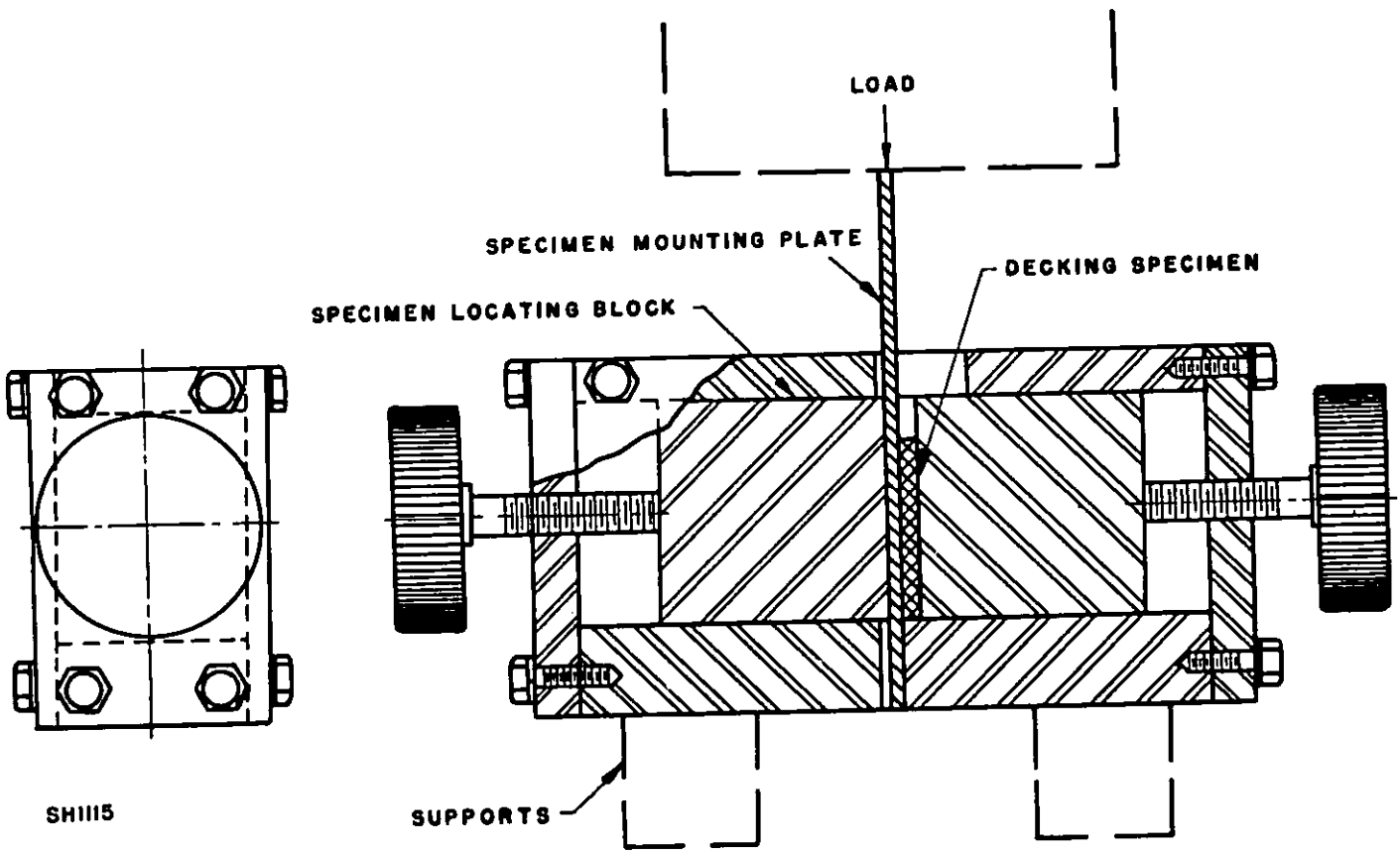


FIGURE 5. Shear test jig for hard-setting deck covering materials.

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